Assignment-9

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1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("NUM.TXT");
    for(int i = 1; i <= 200; i++) {
        fout << i << " ";
    }
    fout.close();
    cout << "Data written successfully.";
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Users\
++ tempCodeRunnerFile.cpp -o t
mpCodeRunnerFile }
Data written successfully.
PS C:\Users\HP\AppData\Local\
```

2. Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

```cpp
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

void countAlphabets() {
    ifstream fin("NOTES.TXT");
    char ch;
    int count = 0;
    while(fin.get(ch)) {
        if(isalpha(ch)) count++;
    }
    cout << "Total alphabets: " << count << endl;
    fin.close();
}

int main() {
    countAlphabets();
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Use
++ tempCodeRunnerFile.cpp
mpCodeRunnerFile }
Data written successfully
PS C:\Users\HP\AppData\Lo
emp\" ; if ($?) { g++ temp
 } ; if ($?) { .\tempCode
Total alphabets: 0
PS C:\Users\HP\AppData\Lo
```

3. Write a program to copy the contents of one file to another.

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream source("source.txt");
    ofstream target("target.txt");
    char ch;
    while(source.get(ch)) {
        target.put(ch);
    }
    source.close();
    target.close();
    cout << "File copied.";
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Users\
++ tempCodeRunnerFile.cpp -o
mpCodeRunnerFile }
File copied.
PS C:\Users\HP\AppData\Local\
```

4. Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    string str;
    cout << "Enter a string: ";
    getline(cin, str);

    int len = str.length();
    cout << "Length: " << len << endl;

    ofstream fout("string_data.txt");
    fout << str;
    fout.close();

    ifstream fin("string_data.txt");
    char ch;
    cout << "Fetching from file: ";
    while(fin.get(ch)) {
        cout << ch;
    }
    fin.close();
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Users\HP\AppDa
++ tempCodeRunnerFile.cpp -o tempCode
mpCodeRunnerFile }
File copied.
PS C:\Users\HP\AppData\Local\Temp> cd
emp\" ; if ($?) { g++ tempCodeRunnerF
 } ; if ($?) { .\tempCodeRunnerFile }
Enter a string: HELLO
Length: 5
Fetching from file: HELLO
PS C:\Users\HP\AppData\Local\Temp>
```

5. Write a program to do the followings:

(a) Create a file A-Z and read 10th character using seekg().

(b) Overwrite 5th number using seekp().

(c) Find file size using tellg().

(d) Read last character using seekg().

(e) Create a file data.txt with multiple lines. Read the file and:

i. Move the pointer to the 10th byte using seekg()

ii. Display the current position using tellg().

iii. Read and display the remaining content.

```cpp
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    fstream file("test.txt", ios::in | ios::out | ios::trunc);
    file << "ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789";

    file.seekg(9, ios::beg);
    char ch10;
    file.get(ch10);
    cout << "10th character: " << ch10 << endl;

    file.seekp(4, ios::beg);
    file.put('*');

    file.seekg(0, ios::end);
    cout << "File size: " << file.tellg() << " bytes" << endl;

    file.seekg(-1, ios::end);
    char last;
    file.get(last);
    cout << "Last character: " << last << endl;

    ofstream fmulti("data.txt");
    fmulti << "Line 1 is here\nLine 2 is here\nLine 3 is here";
    fmulti.close();

    ifstream fin("data.txt");
    fin.seekg(10, ios::beg);
    cout << "Current position after seekg(10): " << fin.tellg() << endl;

    string remaining;
    char c;
    while(fin.get(c)) cout << c;

    file.close();
    fin.close();
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Users\HP\AppData\Loca
++ tempCodeRunnerFile.cpp -o tempCodeRunner
mpCodeRunnerFile }
10th character: J
File size: 36 bytes
Last character: 9
Current position after seekg(10): 10
here
Line 2 is here
Line 3 is here
PS C:\Users\HP\AppData\Local\Temp>
```

6. Write a program to do the followings:

i. Create a file with text "Hello World"

ii. After writing each character, display the current position of the put pointer using tellp().

iii. Replace "World" with "C++" using random access

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    fstream file("hello.txt", ios::in | ios::out | ios::trunc);
    string text = "Hello World";

    for(int i = 0; i < text.length(); i++) {
        file.put(text[i]);
        cout << "Current put pointer: " << file.tellp() << endl;
    }

    file.seekp(6, ios::beg);
    file << "C++  ";

    file.seekg(0, ios::beg);
    string result;
    getline(file, result);
    cout << "Modified content: " << result << endl;

    file.close();
    return 0;
}
```

```
PS C:\Users\HP> cd "C:\Users\HP\AppData\Lo
++ tempCodeRunnerFile.cpp -o tempCodeRunne
mpCodeRunnerFile }
Current put pointer: 1
Current put pointer: 2
Current put pointer: 3
Current put pointer: 4
Current put pointer: 5
Current put pointer: 6
Current put pointer: 7
Current put pointer: 8
Current put pointer: 9
Current put pointer: 10
Current put pointer: 11
Modified content: Hello C++
PS C:\Users\HP\AppData\Local\Temp>
```